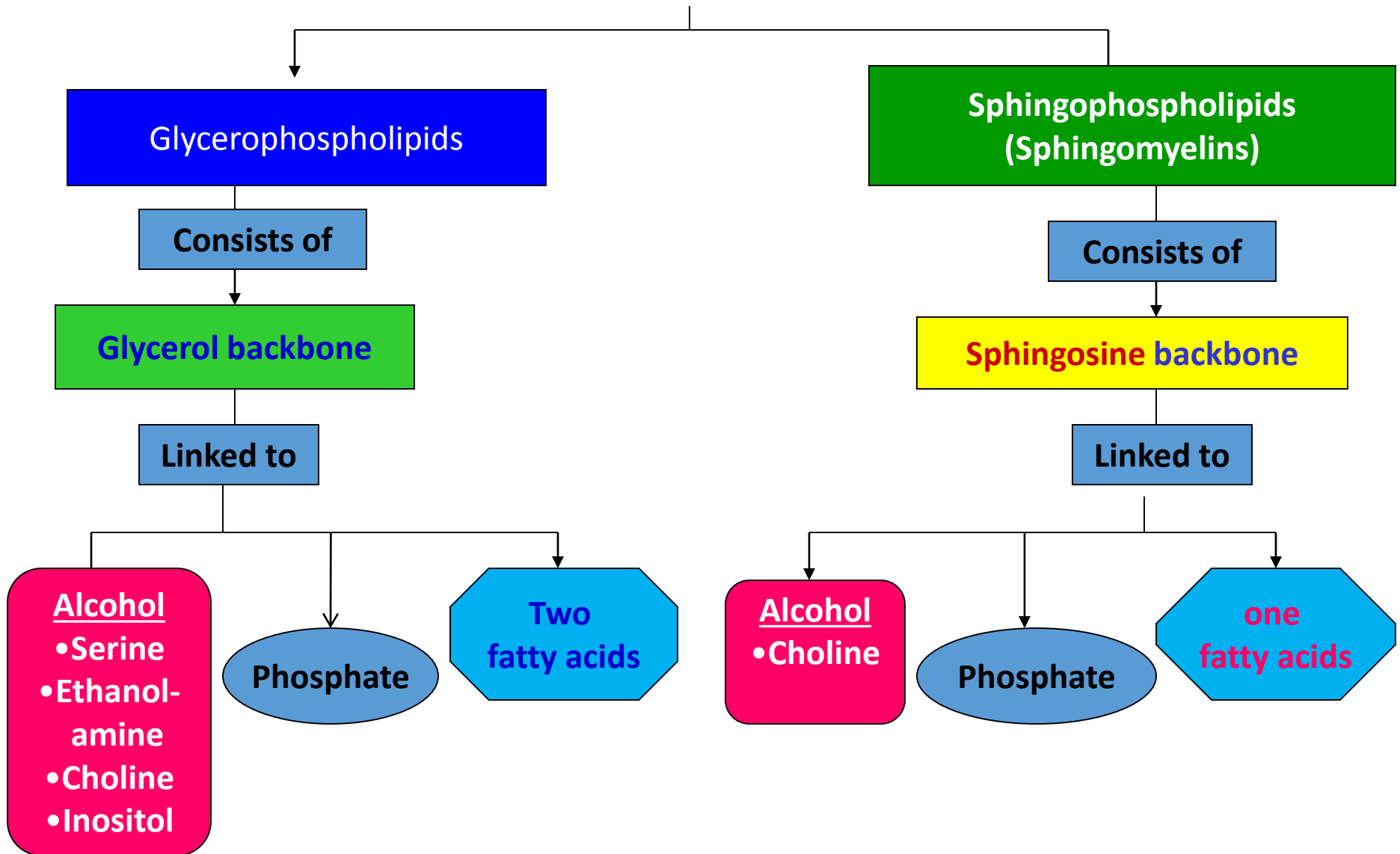


Phospholipids

- They are polar (ionic) compounds composed of an alcohol that is attached by a phosphodiester bridge to either diacylglycerol or sphingosine.
- Classification of Phospholipids
 - Phospholipids are classified into 2 classes:
 - Glycerophospholipids:
 - They have glycerol as a backbone.
 - Sphingophospholipids (Sphingomyelins):
 - They have sphingosine as a backbone.

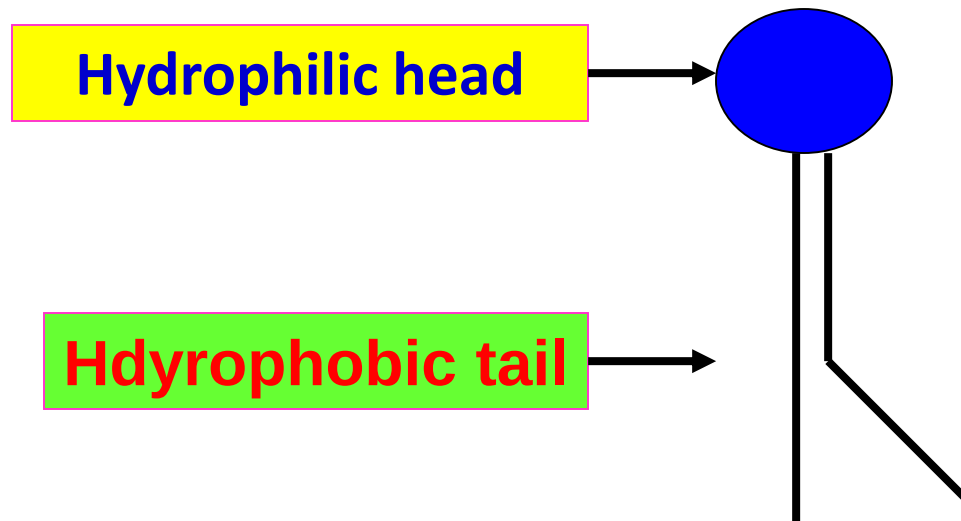
Structure of Phospholipids



Properties of Phospholipids

- They are **amphipathic** molecules i.e. each molecule has a **hydrophilic (polar) head** (formed of the phosphate group and the alcohol group) and a **hydrophobic (non-polar) tail** (formed of glycerol or sphingosine and the hydrocarbon chains of the fatty acids).

Phospholipids are Amphipathic Molecules



Phospholipid molecule

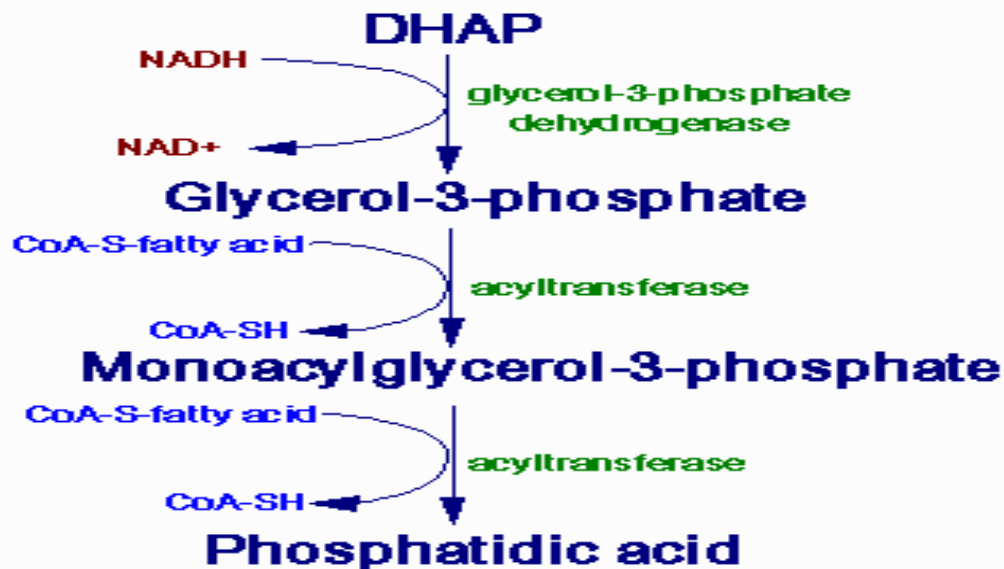
Members of Glycerophospholipids

1. Phosphatidic acid (parent phospholipids).
2. Phosphatidyl choline (Lecithin).
3. Phosphatidyl ethanolamine (cephalin).
4. Phosphatidyl serine.
5. Phosphatidyl inositol.

Phosphatidic acid (parent phospholipids)

- This is the **simplest phospholipid**.
- It **does not occur in good concentration in the tissues**.
- Phosphatidic acid is an intermediate in synthesis of triacylglycerols and phospholipids.

Phosphatidic Acid Synthesis

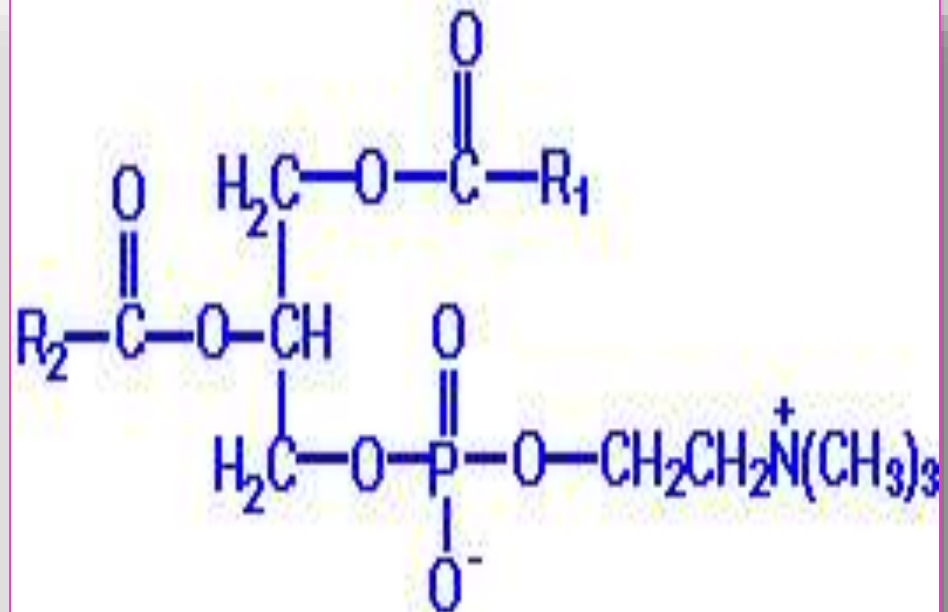


Lecithin

- Lecithin:
- Alcohol + fatty acid + phosphoric acid + choline.
- Present in brain, nervous tissue, sperm and egg yolk.
- Are surface-active agent and help in emulsification of fats.
- Dipalmitoyl lecithin is a lung surfactant (lowers surface tension) prevents the collapse of lung alveoli.

- Absence of dipalmitoyl lecithin in premature infants may produce **respiratory distress syndrome (RDS)** or **hyaline membrane disease**.

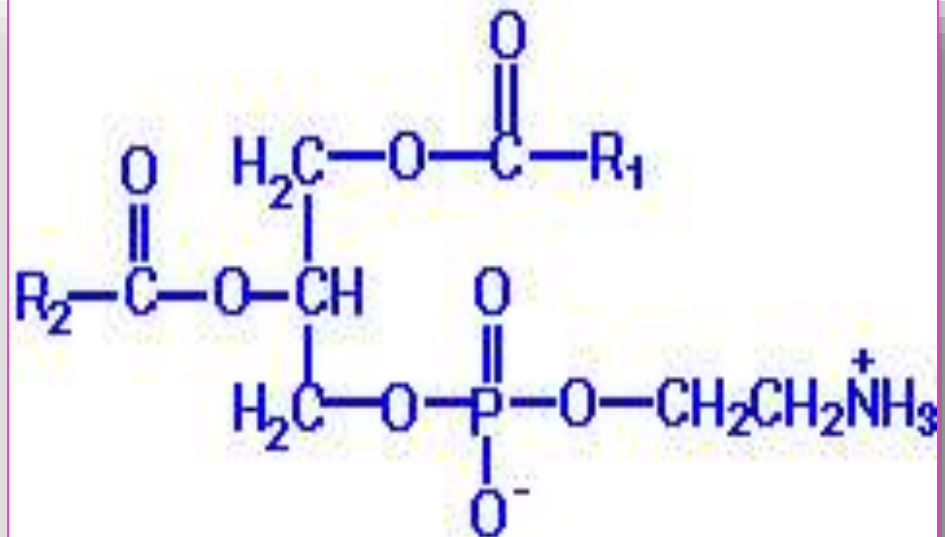
Structure of lecithin



Cephalins

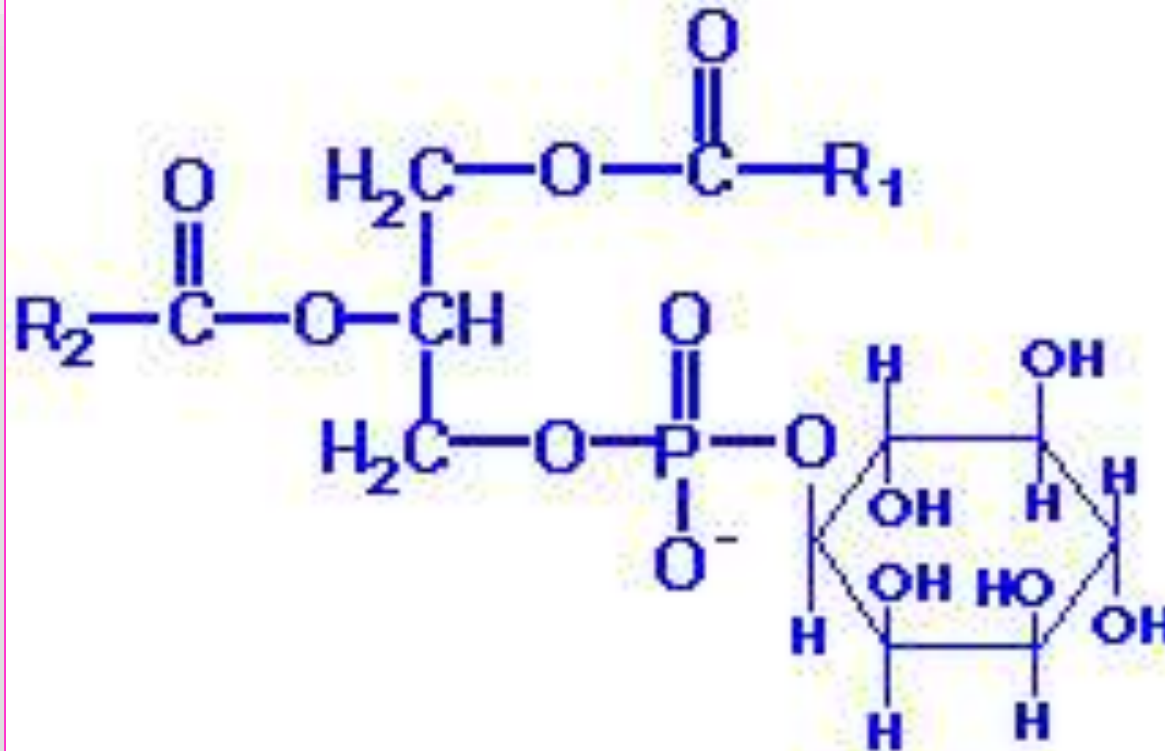
- Cephalins:
- Alcohol+ fatty acid+ phosphoric acid + ethanolamine or serine as a nitrogenous base instead of choline present in lecithin.
- Present in brain, erythrocytes and many other tissues.

Structure of cephalins
(Phosphatidyl ethanolamine)



Phosphatidyl inositol

- **Phosphatidyl Inositol:**
- Phosphatidic acid + inositol (alcohol) instead of a nitrogenous base.
- They are important component of cell membrane.
- The action of certain hormones (e.g. oxytocin, vasopressin) is mediated through phosphatidyl inositol (PI).
- In response to hormonal action, PI is cleaved to diacyl glycerol (DAG) and inositol triphosphate (IP3).
- Both these compounds act as second messenger for hormonal action.

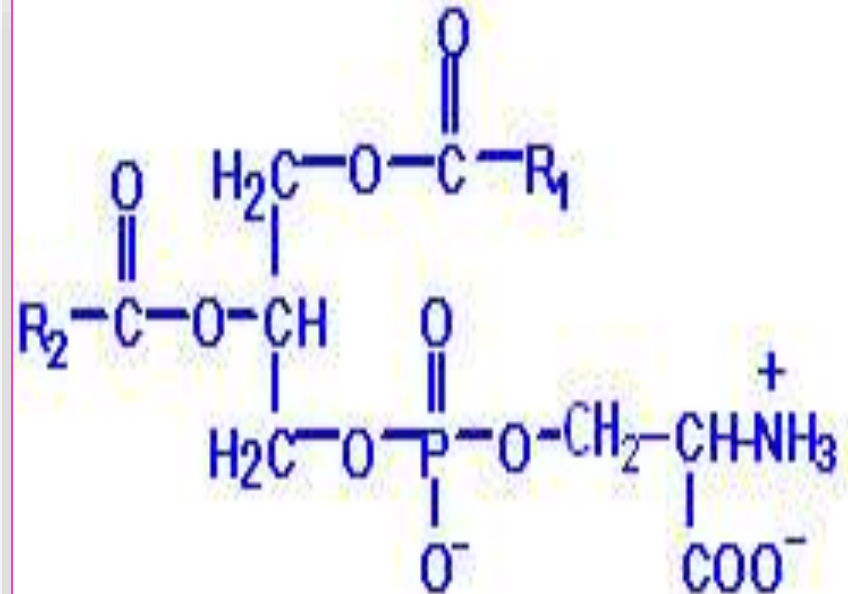


Structure of Phosphatidyl inositol

Phosphatidyl serine

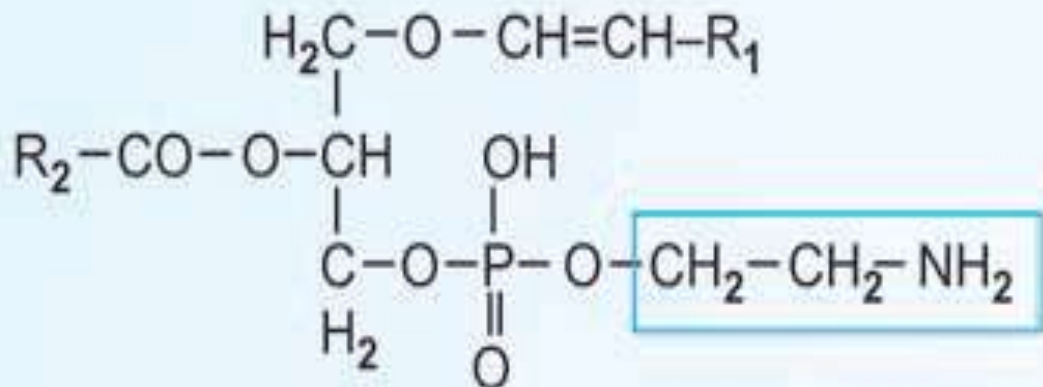
- Phosphatidyl serine:
- The amino acid serine is present in this group of glycerophospholipids.
- Present in brain, nervous tissue and small amount in other tissues.

Structure of Phosphatidyl serine



Plasmalogen

- They differ from lecithin or cephalin in $\alpha 1$ position of glycerol where the fatty acid is replaced by a long chain unsaturated aliphatic aldehyde such as palmitic or stearic aldehyde.
- Plasmalogens are present in large quantities in the skeletal muscle, cardiac muscle and in semen.



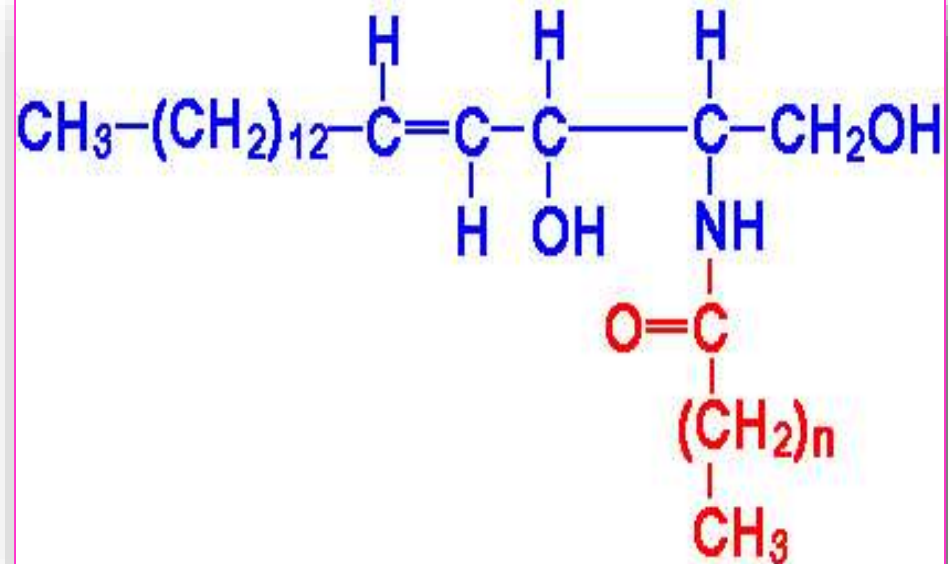
Cardiolipin

- It is diphosphatidyl glycerol.
- Contains two molecules of phosphatidic acid held by glycerol.
- Present in the inner mitochondrial membrane and has antigenic properties.

Ceramide

- Ceramide:
- Formed by the esterification of sphingosine with a fatty acid of high molecular weight.
- Found in white matter of brain and nerves.
- Ceramide is common for all glycolipid and sphingomyelin.

Structure of ceramide



Sphingomyelin

- This is a sphingophospholipid.
- Does not contain glycerol but a unsaturated amino alcohol, i.e. sphingosine.
- Contain a molecule of choline, phosphoric acid and FA
- Sphingomyelin makes up a large part of the myelin sheath.
- These are also present in brain, lungs, nerve and other tissues.
- Deposition of sphingomyelin in liver, lymph nodes, bone marrow and central nervous system results in Neimman-Pick disease due to deficiency of sphingomyelinase enzyme in these tissues.

Glycolipids

- They contain fatty acid, sphingosine (amino alcohol), carbohydrate or carbohydrate derivative.
e.g. Cerebroside, ganglioside
- Cerebrosides:
- Contains fatty acid, sphingosine and a sugar (usually galactose).
- Present in white matter of brain and myelin sheath of nerves.
- Increased in Gaucher's disease in tissues like reticuloendothelial cells of spleen, liver, lymph node and bone.

Gangliosides

- Contains N-acetylneuraminic acid (sialic acid), fatty acid, sphingosine and three molecules of hexoses (glucose or galactose).
- Found in grey matter of the brain
- In Tay Sachs' disease ganglioside level increases.

Lipoproteins

- The different types of lipoproteins are
 - **Chylomicron:**
 - Transports dietary triglyceride and cholesterol esters from intestine to peripheral tissues and liver
 - **Very low density lipoprotein (VLDL):**
 - Transports endogenous triglyceride from liver to extrahepatic tissues.
 - **Low density lipoprotein (LDL):**
 - Transports cholesterol from liver to extrahepatic tissues.

- High density lipoprotein (HDL):

- Transports cholesterol from extrahepatic tissues to the liver in an esterified form.
- Since lipids are water insoluble they are present in the blood in the form of lipoproteins which are water-soluble.
- Composed of triglyceride and cholesterol ester core surrounded by a shell of proteins (also called as Apo proteins), phospholipid and free cholesterol.
- Separated by ultracentrifugation into four distinct groups based on their density (d) and by electrophoresis